Department of Mechanical Engineering

Internal Assessment Question paper

Programme: Common to all branches (UG) **Term:** 07th Oct 2024 – 25th Jan 2025

Course: Design thinking **Course Code:** AECC17

CIE: Test 1 **Semester:** I **Sec:** M,N,O,P,Q,R,S,T,U,V,W & X **Max Marks:** 30

Time: 1Hr **Portions for Test:** Lesson Plan – L1 to L10

Instructions to Candidates:

- Answer any 2 out of 3 questions
- Meaningful Sketches to be drawn with pencil and scale only

Sl. No.		Question's	Marks	Blooms level	Course Outcome
1	a)	Justify why K12 Lab's 6 key design thinking mindsets are important with suitable examples	**10**	**L3**	**CO1**
	b)	Differentiate between NEEDS (FINDINGS) and INSIGHTS with a suitable example	**5**	**L2**	**CO2**
2	a)	Choose suitable example and illustrate each phase in the Design Thinking process	**10**	**L3**	**CO1**
	b)	Briefly explain i. Immersion ii. Interviews	**5**	**L2**	**CO2**
3	a)	What is Design Thinking? How does it work?	**5**	**L2**	**CO1**
	b)	Write 10 interview questions to ask ambulance drivers pertaining to problems faced while ferrying patients to the hospital	**10**	**L3**	**CO2**

MOBILE PHONES ARE BANNED IN THE EXAM HALL

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Scheme of Evaluation

Instructions to Candidates:

- Answer any 2 out of 3 questions
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Sl. No.		Question's	Marks
1	a)	K12 Lab's six key design thinking mindsets—Creative Confidence, Bias Toward Action, Human-Centered, Show Don't Tell, Radical Collaboration, and Mindful of Process—are vital for fostering innovation and problem-solving in education and beyond. Here's why each is important, with examples: 1. Creative Confidence **Importance:** Encourages individuals to trust their ability to think creatively and solve problems. **Example:** A teacher designs an interactive science experiment despite limited resources, inspiring students to explore concepts hands-on. 2. Bias Toward Action **Importance:** Focuses on learning by doing rather than over-planning, which fosters experimentation and rapid prototyping. **Example:** Students build a simple prototype of a water filtration system instead of spending weeks researching designs, learning through trial and error. 3. Human-Centered **Importance:** Emphasizes empathy and understanding the needs of the people you are designing for. **Example:** A school redesigns its library layout after interviewing students and realizing they prefer flexible seating for group projects. 4. Show Don't Tell **Importance:** Advocates for visual communication to convey ideas effectively, making abstract concepts tangible.	**10**

Example: Instead of explaining a new app idea verbally, a student creates a storyboard to show how users would interact with it.

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5. Radical Collaboration

Importance: Promotes diverse teamwork, which leads to richer ideas and better solutions.
Example: A team of teachers, parents, and students collaborate to redesign the school's playground, ensuring all perspectives are considered.

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6. Mindful of Process

Importance: Encourages staying aware of the design process to ensure it remains iterative and goal-oriented.
Example: A teacher facilitates a project on reducing plastic waste, guiding students to revisit and refine their ideas after feedback sessions.

Conclusion

By embracing these mindsets, individuals and teams can navigate challenges innovatively and empathetically, creating impactful solutions tailored to real-world needs. Together, these principles transform how problems are approached, making them indispensable in design thinking frameworks

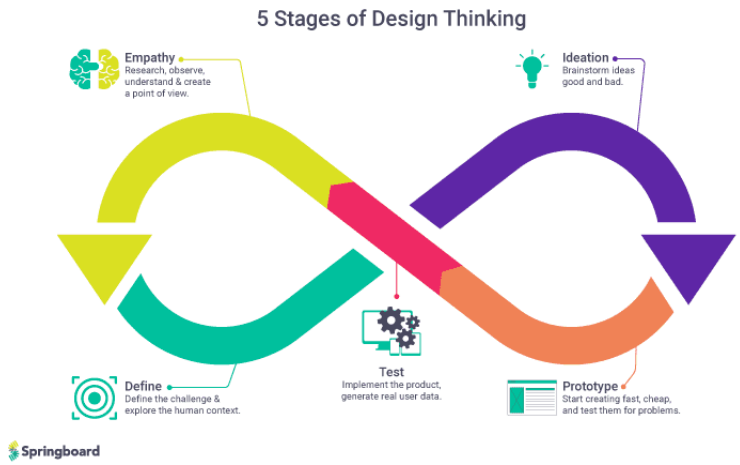
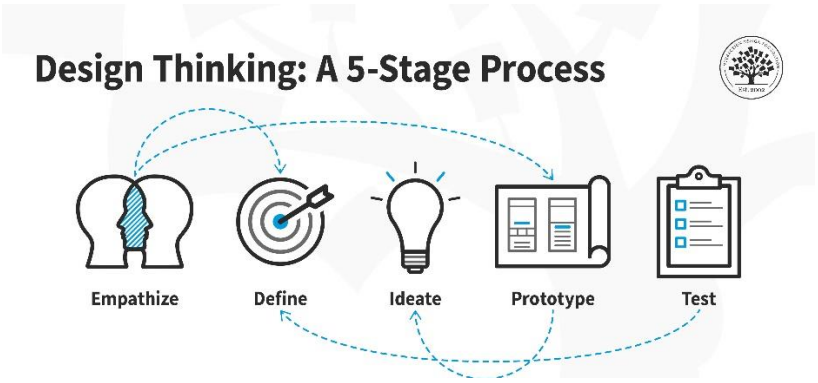
b) NEEDS (FINDINGS): User's emotional or physical necessities and desires that come in form of verbs.

INSIGHTS: Remarkable realizations that could be leveraged to better respond to a design challenge. – **2 marks**

Differentiating with example – **3 marks**

FINDING	INSIGHT
Community members drive to other locations for food and entertainment	Community members drive to other locations for food and entertainment because there are not many options in their area
Community members like to visit places of relaxation during weekends	Community members like to visit places of relaxation during weekends as they are exhausted by work/ travel
Very few Community members have been vaccinated	Very few Community members have been vaccinated because of fear, as there was fake news spreading in the area
Community members are traveling to stay in other locations during covid lockdown	Community members are traveling to stay in other locations during covid lockdown as there are very few hospitals in this area

10

2	a)	**Mentioning the Process – 2 marks**  **Example – 2 marks** **Briefly explaining the Process – 6 marks**	**10**
	b)	Briefly explaining the below mentioned with examples - **5 marks** i. Immersion ii. Interviews	**5**
3	a)	Design thinking is a non-linear, iterative process that teams use to understand users, challenge assumptions, redefine problems and create innovative solutions to prototype and test. Involving five phases—Empathize, Define, Ideate, Prototype and Test—it is most useful to tackle problems that are ill-defined or unknown – **3 marks** Mention the process below – **2 marks**  	**5**

	b)	Listing 10 logical questions to ask ambulance drivers regarding problems faced while ferrying patients to the hospital – **1x10 = 10 marks**	**10**